

PAYEL BHATTACHARJEE

Tucson, AZ, USA

✉ payelb@arizona.edu ✉ payelbhattacharjee09@gmail.com  LinkedIn  GitHub  Website  Google Scholar

EDUCATION

University of Arizona, Tucson, AZ, USA
Ph.D. in Electrical and Computer Engineering
Advisor: Dr. Ravi Tandon

Aug 2022 – Present
GPA: 3.79/4.00

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, India
B.Tech. in Electronics & Telecommunication Engineering

2017 – 2021
GPA: 9.43/10.0

RESEARCH INTERESTS

Trustworthy Machine Learning; Large Language Model Alignment and Evaluation; Reward Modeling and Preference Learning; Differential Privacy; Causal Inference and Causal Graph Discovery; Efficient LLM Inference; Privacy-Preserving Machine Learning.

RESEARCH EXPERIENCE

University of Arizona
Graduate Research Associate — Advisor: Dr. Ravi Tandon

Tucson, AZ
Aug 2022 – Present

- Developed **adaptive differential privacy mechanisms for causal graph discovery** that allocate privacy budget according to the informativeness and difficulty of causal tests, improving graph recovery under fixed privacy constraints.
- Developed **privacy-preserving fine-tuning and alignment methods for large language models**, enabling models to learn from sensitive textual and preference data while providing formal differential privacy guarantees.
- Proposed **margin- and semantic-aware reward modeling** methods that identify difficult or uncertain preference comparisons and selectively augment them, improving learning efficiency under limited and noisy human-preference data.
- Developed a **bandwidth-efficient speculative decoding framework for edge–cloud LLM inference** that uses conformal prediction to adaptively sparsify communication while maintaining reliable token prediction.
- Developed **privacy-preserving learning methods for sensitive healthcare data**, including differentially private LLM-based radiology report classification and multimodal diagnostic learning.
- Investigated **reliable evaluation of aligned language models** using reward-model discrimination, preference-based evaluation, and LLM judges, with emphasis on difficult comparisons, evaluator disagreement, and robustness of alignment metrics.

PUBLICATIONS

Large Language Models, Alignment, and Efficient Inference

1. **P. Bhattacharjee**, R. Tandon. *Adaptive Multi-Value Control in LLMs via Causal Activation Steering*. arXiv preprint, 2026. **Under Review**.
2. **P. Bhattacharjee**, et al. *Conformal Speculative Speculative Decoding*. Accepted for presentation in ASILOMAR 2026 Conference (full paper forthcoming).
3. **P. Bhattacharjee**, et al. *MARS: Margin and Semantic-Aware Data Augmentation for Reward Modeling*. arXiv preprint, 2026. **Under Review**.
4. N. Teku, F. Tian, **P. Bhattacharjee**, et al. *PROPS: Progressively Private Self-alignment of LLMs*. *Transactions on Machine Learning Research (TMLR)*, 2025.
5. **P. Bhattacharjee***, F. Tian*, M. Zhong*, et al. *Conformal Sparsification for Bandwidth-Efficient Speculative Decoding*. *NeurIPS AI4NextG Workshop*, 2025.

Privacy-Preserving Machine Learning and Causal Inference

4. **P. Bhattacharjee**, et al. *Learning to Diagnose Privately: DP-Powered LLMs for Radiology Report Classification*. *IEEE Access*, 2026.

5. F. Tian, **P. Bhattacharjee**, et al. *STAMP: Selective Task-Aware Mechanism for Text Privacy*. *European Chapter of the Association for Computational Linguistics (EACL)*, 2026.
6. **P. Bhattacharjee**, R. Tandon. *CURATE: Scaling Up Differentially Private Causal Graph Discovery*. *Entropy*, 2024.
7. **P. Bhattacharjee**, R. Tandon. *Adaptive Privacy for Differentially Private Causal Graph Discovery*. *IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2024.
8. Z. Guan, Z. Zhao, F. Tian, D. Nguyen, **P. Bhattacharjee**, et al. *A Framework for Multi-source Privacy Preserving Epidemic Analysis*. arXiv preprint, 2025.

Other Publications

9. **P. Bhattacharjee**, et al. *Numerical Modelling and Performance Evaluation of SnS-Based Heterojunction Solar Cell with p^+ -SnS BSF Layer*. *Optical and Quantum Electronics*, 2022.
10. **P. Bhattacharjee**, et al. *Sleep and Sedentary Behavior Analysis from Physiological Signals Using Machine Learning*. *IEEE International Conference on Industrial and Information Systems (ICIMIA)*, 2020.

*Equal contribution.

TEACHING EXPERIENCE

University of Arizona

Graduate Teaching Assistant, ECE 372

Tucson, AZ

Aug 2022 – May 2023

I served as a Teaching Assistant for the ECE372 course, managing a class of 60 students. In this role, I oversaw laboratory sessions, guided students through experiments, and provided hands-on support in debugging and refining their implemented code. Additionally, I coordinated classroom activities, ensured smooth execution of lab exercises, and facilitated effective learning by addressing individual and group challenges. This experience strengthened my skills in both technical mentorship and classroom management.

AWARDS & HONORS

GPSC Travel Grant, University of Arizona

2024, 2025

People's Choice Award, ECE Graduate Poster Symposium, University of Arizona

2024

KIIT Merit Scholarship, KIIT University

2018

ACADEMIC SERVICE

Officer, Electrical and Computer Engineering Graduate Student Association, University of Arizona 2024 – Present

TECHNICAL SKILLS

Machine Learning: Large Language Models, Transformers, RLHF, Reward Modeling, Preference Learning, Differential Privacy, Causal Inference, Causal Discovery, Conformal Prediction, Federated Learning, Speculative Decoding.

Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA.

Programming: Python, C/C++, R, MATLAB.

Computing: GPU Computing, HPC Systems, Multi-GPU Training.

INDUSTRY EXPERIENCE

Bosch Global Software Technologies (BGSW)

India

Associate Software Engineer

Jul 2021 – Jun 2022

- Developed Python-based software and automation tools for Advanced Driver Assistance Systems (ADAS) data-analysis workflows.
- Developed data-processing and machine-learning solutions for automotive projects involving clients in the United States and Europe.

Bosch Global Software Technologies (BGSW)

India

Project Intern

Mar 2021 – Jun 2021

- Developed Python and machine-learning tools for automated data analysis and engineering workflows.